

Conic Sections

Parabola, Ellipse and Hyperbola

Focus–Directrix Definitions, Standard Forms, Formulae and Examples

Based on the material in Sections 10.5 and 11.8

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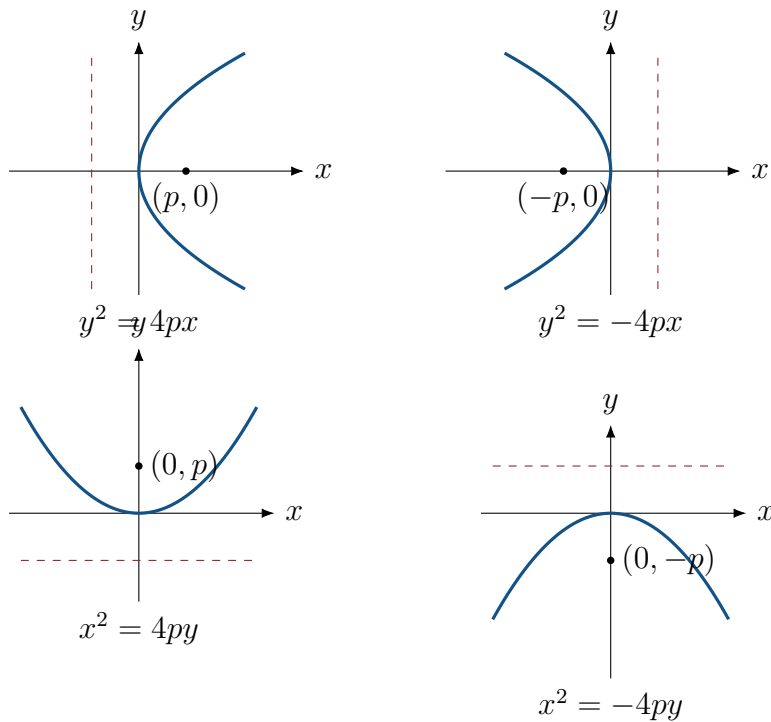
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1.2 Parabolas in Standard Position



Necessary Formulae for Parabolas in Standard Position ($p > 0$)

Description	$y^2 = 4px$	$y^2 = -4px$	$x^2 = 4py$	$x^2 = -4py$
Vertex	$(0, 0)$	$(0, 0)$	$(0, 0)$	$(0, 0)$
Focus	$(p, 0)$	$(-p, 0)$	$(0, p)$	$(0, -p)$
Directrix	$x = -p$	$x = p$	$y = -p$	$y = p$
Axis	$y = 0$	$y = 0$	$x = 0$	$x = 0$
Latus rectum line	$x = p$	$x = -p$	$y = p$	$y = -p$
Length of latus rectum	$4p$	$4p$	$4p$	$4p$
Opening	right	left	upward	downward

Example 1: Focus, Directrix, and Graph

Find the focus and directrix of the parabola

$$y^2 + 10x = 0,$$

and sketch its graph.

Answer

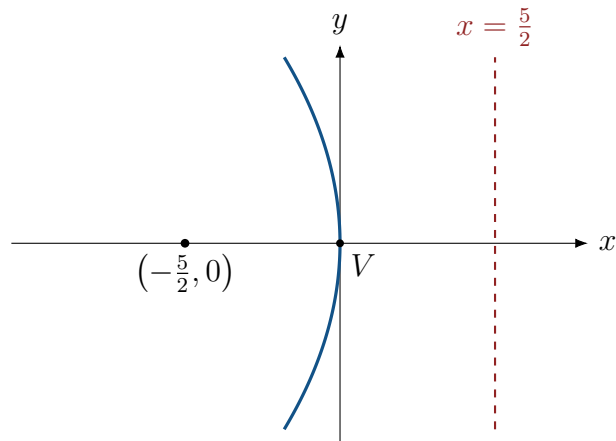
$$y^2 + 10x = 0,$$

$$y^2 = -10x = 4 \left(-\frac{5}{2} \right) x.$$

Thus $p = -\frac{5}{2}$. Therefore,

$$\text{focus} = \left(-\frac{5}{2}, 0\right), \quad \text{directrix} : x = \frac{5}{2},$$

and the parabola opens to the left.



1.3 Parabolas with Vertex (α, β)

The standard equations are

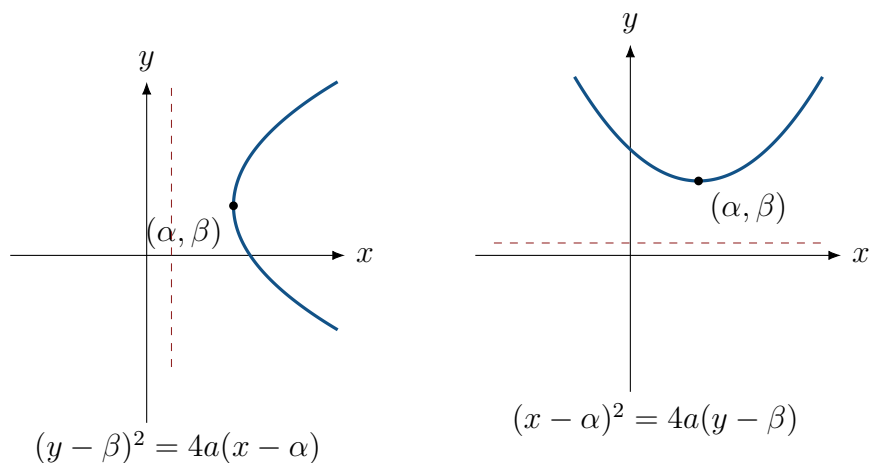
$$(y - \beta)^2 = 4a(x - \alpha), \quad (x - \alpha)^2 = 4a(y - \beta).$$

They are obtained from $Y^2 = 4aX$ and $X^2 = 4aY$ by putting

$$X = x - \alpha, \quad Y = y - \beta.$$

Necessary Formulae for Translated Parabolas

Description	$(y - \beta)^2 = 4a(x - \alpha)$	$(x - \alpha)^2 = 4a(y - \beta)$
Vertex	(α, β)	(α, β)
Focus	$(\alpha + a, \beta)$	$(\alpha, \beta + a)$
Directrix	$x = \alpha - a$	$y = \beta - a$
Axis	$y = \beta$	$x = \alpha$
Latus rectum line	$x = \alpha + a$	$y = \beta + a$
Length of latus rectum	$4 a $	$4 a $
Opening	right if $a > 0$, left if $a < 0$	up if $a > 0$, down if $a < 0$



Example 2: Describe the Parabola

Describe the graph of

$$y^2 - 8x - 6y - 23 = 0.$$

Answer

Complete the square in y :

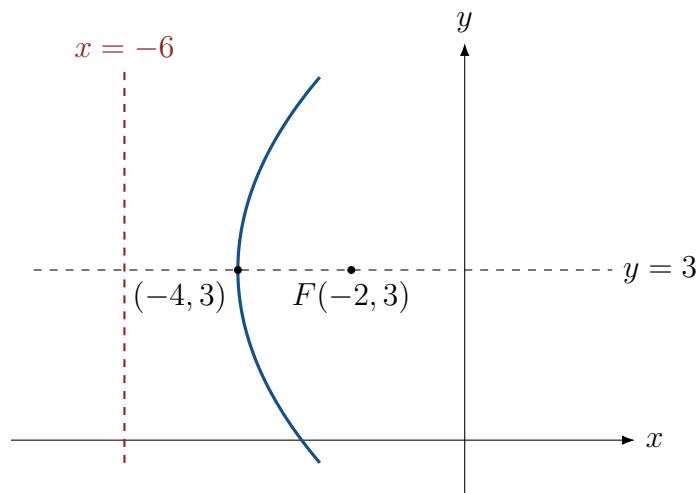
$$\begin{aligned} y^2 - 6y - 8x - 23 &= 0, \\ y^2 - 6y + 9 &= 8x + 32, \\ (y - 3)^2 &= 8(x + 4) = 4(2)(x - (-4)). \end{aligned}$$

Hence

$$\text{vertex} = (-4, 3), \quad a = 2, \quad \text{focus} = (-2, 3), \quad \text{directrix} : x = -6,$$

with axis $y = 3$ and latus rectum line $x = -2$.

This is the algebraically correct sign completion of the handwritten equation.



2 Ellipse

Definition of an Ellipse

An **ellipse** may be described equivalently in either of the following ways.

Two-focus definition: It is the set of all points P in a plane for which the sum of the distances from two fixed points F_1 and F_2 is constant:

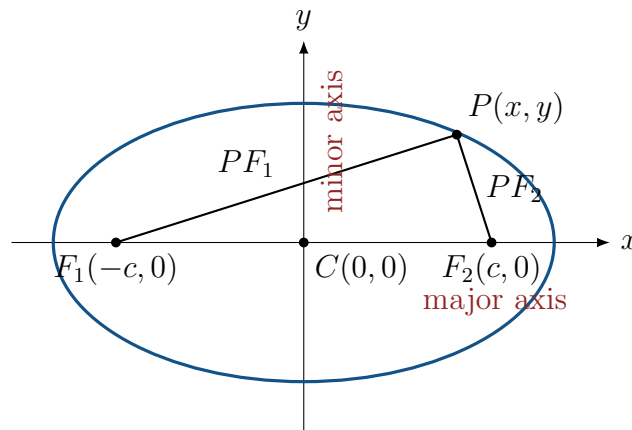
$$PF_1 + PF_2 = 2a.$$

The fixed points are called the **foci**. Their midpoint is the **centre**. The longer axis is the **major axis**, and the shorter axis is the **minor axis**.

Focus–directrix definition: Let S be a fixed point (the focus), let ℓ be a fixed line (the directrix), and let M be the foot of the perpendicular from P to ℓ . An ellipse is the locus of points P satisfying

$$SP = e PM, \quad 0 < e < 1.$$

The constant e is called the **eccentricity**. Because $e < 1$, every point of the ellipse is closer to the focus than to the corresponding directrix.



2.1 Focus–Directrix Form and Derivation of the Standard Equation

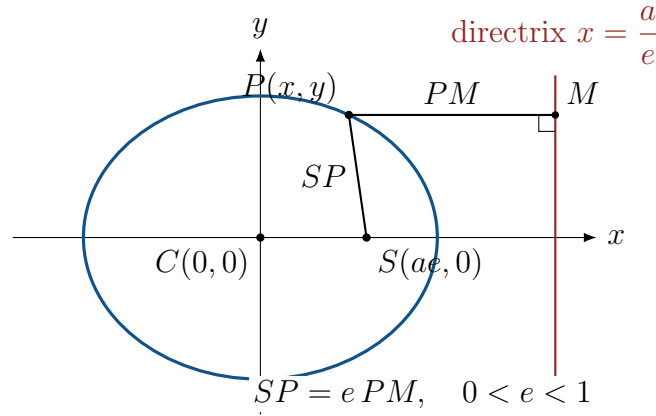
Consider a horizontal ellipse centred at the origin. Take the right-hand focus

$$S(ae, 0)$$

and its corresponding directrix

$$x = \frac{a}{e}, \quad 0 < e < 1.$$

For a point $P(x, y)$ on the ellipse, let M be the foot of the perpendicular from P to the directrix. Then $PM = \frac{a}{e} - x$, and the defining relation is $SP = e PM$.



Using the distance formula,

$$SP = e PM,$$

$$\sqrt{(x - ae)^2 + y^2} = e \left(\frac{a}{e} - x \right) = a - ex.$$

Squaring both sides and simplifying,

$$\begin{aligned} (x - ae)^2 + y^2 &= (a - ex)^2, \\ x^2 - 2aex + a^2e^2 + y^2 &= a^2 - 2aex + e^2x^2, \\ (1 - e^2)x^2 + y^2 &= a^2(1 - e^2). \end{aligned}$$

Since $0 < e < 1$, define

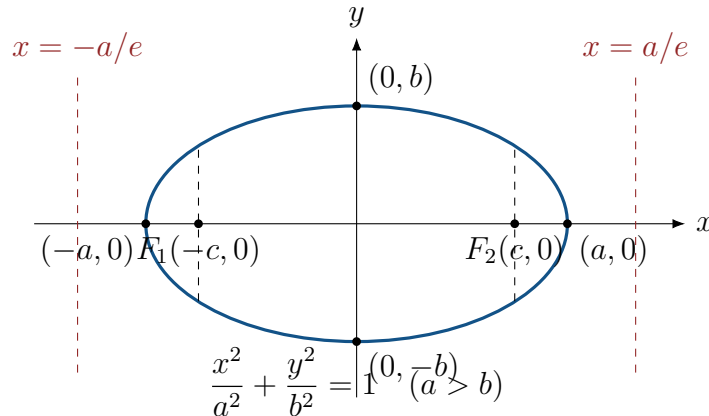
$$b^2 = a^2(1 - e^2) > 0.$$

Dividing by $a^2(1 - e^2) = b^2$ gives

$$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1}, \quad a > b > 0.$$

Also,

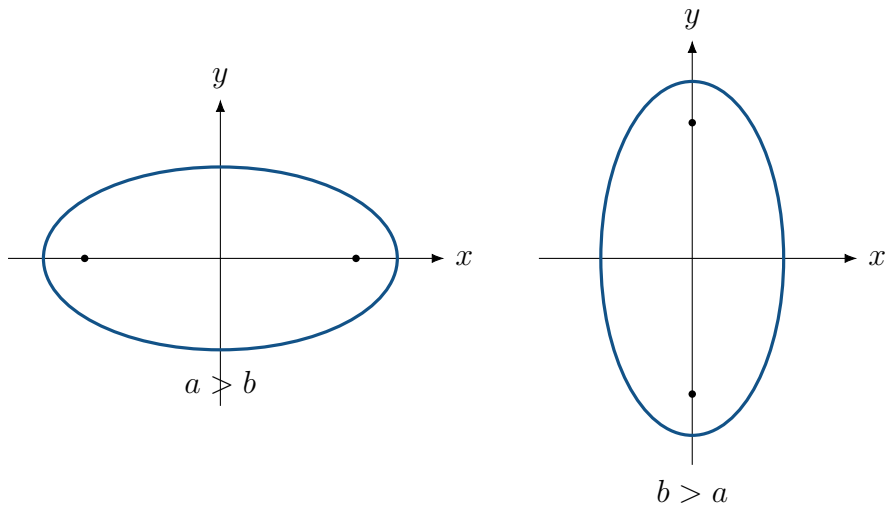
$$c = ae, \quad c^2 = a^2 - b^2, \quad e = \frac{c}{a} = \sqrt{\frac{a^2 - b^2}{a^2}}.$$



2.2 Horizontal and Vertical Ellipses Centred at the Origin

Necessary Formulae for Ellipses Centred at $(0, 0)$

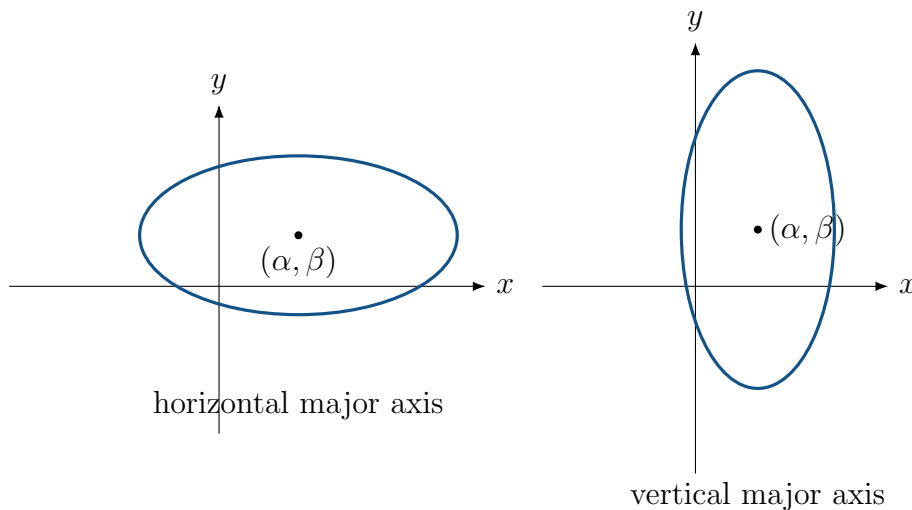
Description	Horizontal major axis	Vertical major axis
Equation	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, b > a$
Centre	$(0, 0)$	$(0, 0)$
Eccentricity	$e = \frac{\sqrt{a^2 - b^2}}{a}$	$e = \frac{\sqrt{b^2 - a^2}}{b}$
Vertices	$(\pm a, 0)$	$(0, \pm b)$
Co-vertices	$(0, \pm b)$	$(\pm a, 0)$
Foci	$(\pm ae, 0)$	$(0, \pm be)$
Directrices	$x = \pm \frac{a}{e}$	$y = \pm \frac{b}{e}$
Latus rectum lines	$x = \pm ae$	$y = \pm be$
Major axis	$y = 0$	$x = 0$
Minor axis	$x = 0$	$y = 0$
Area	πab	πab



2.3 Ellipses with Centre (α, β)

Necessary Formulae for Translated Ellipses

Description	Horizontal major axis ($a > b$)	Vertical major axis ($b > a$)
Equation	$\frac{(x - \alpha)^2}{a^2} + \frac{(y - \beta)^2}{b^2} = 1$	$\frac{(x - \alpha)^2}{a^2} + \frac{(y - \beta)^2}{b^2} = 1$
Centre	(α, β)	(α, β)
Eccentricity	$e = \frac{\sqrt{a^2 - b^2}}{a}$	$e = \frac{\sqrt{b^2 - a^2}}{b}$
Vertices	$(\alpha \pm a, \beta)$	$(\alpha, \beta \pm b)$
Co-vertices	$(\alpha, \beta \pm b)$	$(\alpha \pm a, \beta)$
Foci	$(\alpha \pm ae, \beta)$	$(\alpha, \beta \pm be)$
Directrices	$x = \alpha \pm \frac{a}{e}$	$y = \beta \pm \frac{b}{e}$
Major axis	$y = \beta$	$x = \alpha$
Minor axis	$x = \alpha$	$y = \beta$
Latus rectum lines	$x = \alpha \pm ae$	$y = \beta \pm be$



Example 3: Ellipse Centred at the Origin

Find the eccentricity, centre, foci, vertices, equations of the latus recta, equations of the directrices, and the major and minor axes of

$$16x^2 + 25y^2 = 400.$$

Answer

Divide by 400:

$$\frac{x^2}{25} + \frac{y^2}{16} = 1.$$

Thus $a = 5$, $b = 4$, and the major axis is horizontal. Therefore,

$$e = \sqrt{\frac{a^2 - b^2}{a^2}} = \sqrt{\frac{25 - 16}{25}} = \frac{3}{5},$$

$$\text{centre} = (0, 0),$$

$$\text{foci} = (\pm ae, 0) = (\pm 3, 0),$$

$$\text{vertices} = (\pm 5, 0),$$

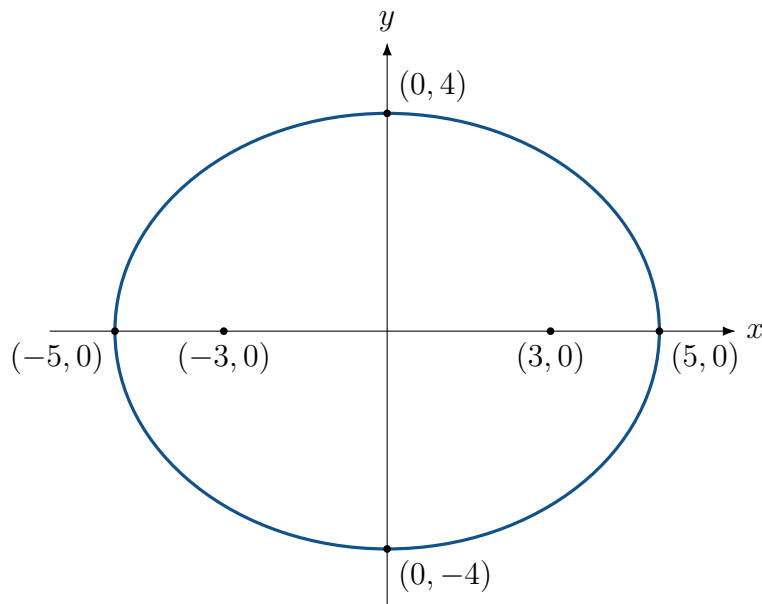
$$\text{co-vertices} = (0, \pm 4),$$

$$\text{latus rectum lines : } x = \pm 3,$$

$$\text{directrices : } x = \pm \frac{a}{e} = \pm \frac{25}{3},$$

$$\text{major axis : } y = 0,$$

$$\text{minor axis : } x = 0.$$



Practice from the Notes

Repeat the same analysis for

$$5x^2 + 4y^2 = 20.$$

Example 4: A Translated Ellipse

Find the vertices and foci, find the equations of the directrices, and sketch the graph of

$$9x^2 - 54x + y^2 + 2y + 46 = 0.$$

Answer

Complete the squares:

$$\begin{aligned} 9(x^2 - 6x) + (y^2 + 2y) + 46 &= 0, \\ 9((x - 3)^2 - 9) + ((y + 1)^2 - 1) + 46 &= 0, \\ 9(x - 3)^2 + (y + 1)^2 &= 36, \\ \frac{(x - 3)^2}{4} + \frac{(y + 1)^2}{36} &= 1. \end{aligned}$$

The major axis is vertical. Here the horizontal semi-axis is $a = 2$ and the vertical semi-major axis is $b = 6$. Thus

$$\text{centre} = (3, -1),$$

$$e = \sqrt{\frac{b^2 - a^2}{b^2}} = \sqrt{\frac{36 - 4}{36}} = \frac{2\sqrt{2}}{3},$$

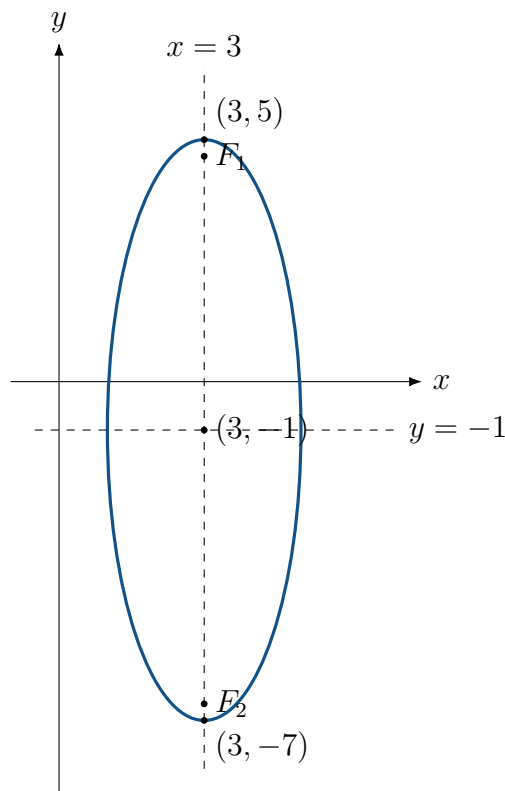
$$c = be = 4\sqrt{2},$$

$$\text{vertices} = (3, -1 \pm 6) = (3, 5), (3, -7),$$

$$\text{co-vertices} = (3 \pm 2, -1) = (5, -1), (1, -1),$$

$$\text{foci} = (3, -1 \pm 4\sqrt{2}),$$

$$\text{directrices : } y = -1 \pm \frac{b}{e} = -1 \pm \frac{9}{\sqrt{2}}.$$



3 Hyperbola

Definition of a Hyperbola

A **hyperbola** may be described equivalently in either of the following ways.

Two-focus definition: It is the set of all points P in a plane for which the absolute difference of the distances from two fixed points F_1 and F_2 is constant:

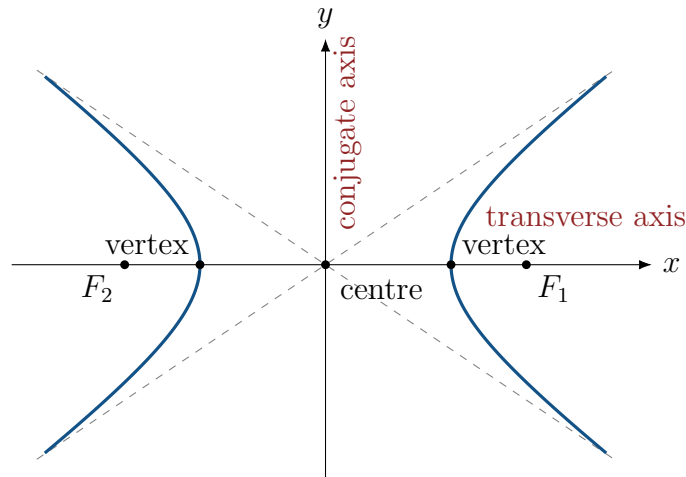
$$|PF_1 - PF_2| = 2a.$$

The fixed points are the **foci**. The line through the foci is the **transverse axis**; the perpendicular line through the centre is the **conjugate axis**.

Focus–directrix definition: Let S be a fixed point (the focus), let ℓ be a fixed line (the directrix), and let M be the foot of the perpendicular from P to ℓ . A hyperbola is the locus of points P satisfying

$$SP = e PM, \quad e > 1.$$

The constant e is the **eccentricity**. Because $e > 1$, every point on the corresponding branch is farther from the focus than from the corresponding directrix.



3.1 Focus–Directrix Form and Derivation of the Standard Equation

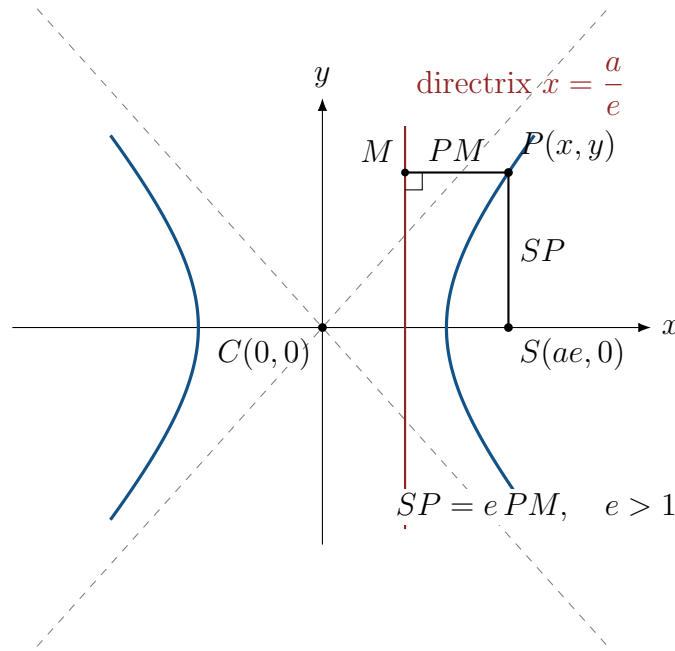
Consider the right branch of a horizontal hyperbola centred at the origin. Take the right-hand focus

$$S(ae, 0)$$

and its corresponding directrix

$$x = \frac{a}{e}, \quad e > 1.$$

For a point $P(x, y)$ on the right branch, let M be the foot of the perpendicular from P to the directrix. Since $x > \frac{a}{e}$, we have $PM = x - \frac{a}{e}$, and the defining relation is $SP = e PM$.



Using the distance formula,

$$SP = e PM,$$

$$\sqrt{(x - ae)^2 + y^2} = e \left(x - \frac{a}{e} \right) = ex - a.$$

Squaring both sides and simplifying,

$$\begin{aligned} (x - ae)^2 + y^2 &= (ex - a)^2, \\ x^2 - 2aex + a^2e^2 + y^2 &= e^2x^2 - 2aex + a^2, \\ (e^2 - 1)x^2 - y^2 &= a^2(e^2 - 1). \end{aligned}$$

Since $e > 1$, define

$$b^2 = a^2(e^2 - 1) > 0.$$

Dividing by $a^2(e^2 - 1) = b^2$ gives

$$\boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1}.$$

Moreover,

$$c = ae, \quad c^2 = a^2 + b^2, \quad e = \frac{c}{a} = \sqrt{\frac{a^2 + b^2}{a^2}} > 1.$$

The left branch is obtained from the left focus $(-ae, 0)$ and the corresponding directrix $x = -a/e$.

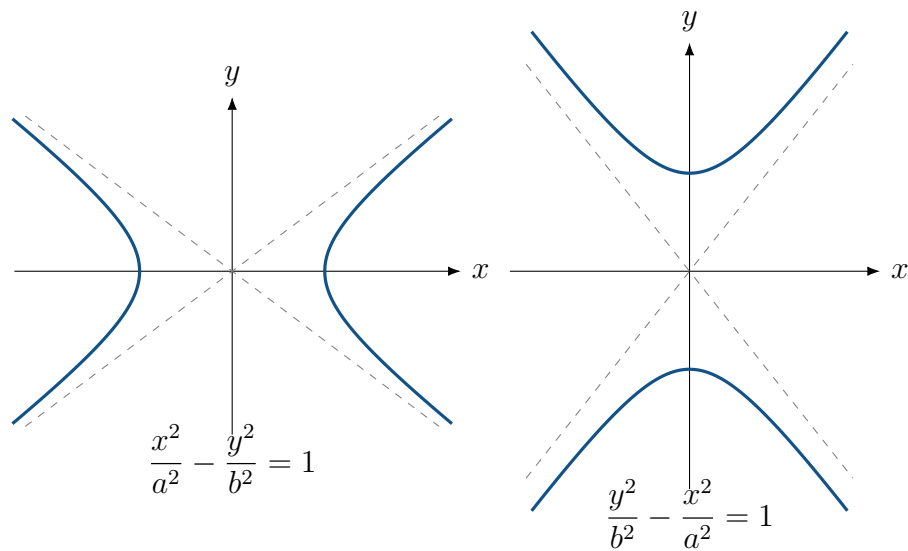
3.2 Standard Forms Centred at the Origin

For a horizontal hyperbola,

$$\boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1}, \quad c^2 = a^2 + b^2, \quad c = ae, \quad e = \frac{c}{a} = \sqrt{\frac{a^2 + b^2}{a^2}} > 1.$$

For a vertical hyperbola, using the notation of the handwritten notes,

$$\boxed{\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1}, \quad c^2 = a^2 + b^2, \quad c = be, \quad e = \frac{c}{b} = \sqrt{\frac{a^2 + b^2}{b^2}} > 1.$$



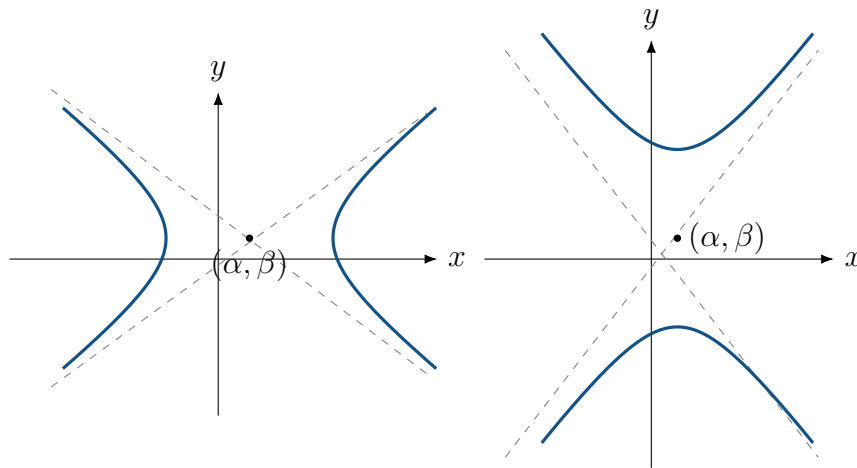
Necessary Formulae for Hyperbolas Centred at $(0, 0)$

Description	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$
Centre	$(0, 0)$	$(0, 0)$
Eccentricity	$e = \frac{\sqrt{a^2 + b^2}}{a}$	$e = \frac{\sqrt{a^2 + b^2}}{b}$
Foci	$(\pm ae, 0)$	$(0, \pm be)$
Vertices	$(\pm a, 0)$	$(0, \pm b)$
Directrices	$x = \pm \frac{a}{e}$	$y = \pm \frac{b}{e}$
Transverse axis	$y = 0$	$x = 0$
Conjugate axis	$x = 0$	$y = 0$
Asymptotes	$y = \pm \frac{b}{a}x$	$y = \pm \frac{b}{a}x$
Latus rectum lines	$x = \pm ae$	$y = \pm be$

3.3 Hyperbolas with Centre (α, β)

Necessary Formulae for Translated Hyperbolas

Description	Horizontal transverse axis	Vertical transverse axis
Equation	$\frac{(x - \alpha)^2}{a^2} - \frac{(y - \beta)^2}{b^2} = 1$	$\frac{(y - \beta)^2}{b^2} - \frac{(x - \alpha)^2}{a^2} = 1$
Centre	(α, β)	(α, β)
Eccentricity	$e = \frac{\sqrt{a^2 + b^2}}{a}$	$e = \frac{\sqrt{a^2 + b^2}}{b}$
Foci	$(\alpha \pm ae, \beta)$	$(\alpha, \beta \pm be)$
Vertices	$(\alpha \pm a, \beta)$	$(\alpha, \beta \pm b)$
Directrices	$x = \alpha \pm \frac{a}{e}$	$y = \beta \pm \frac{b}{e}$
Transverse axis	$y = \beta$	$x = \alpha$
Conjugate axis	$x = \alpha$	$y = \beta$
Asymptotes	$y - \beta = \pm \frac{b}{a}(x - \alpha)$	$y - \beta = \pm \frac{b}{a}(x - \alpha)$



Example 5: A Vertical Hyperbola

Find the eccentricity, the foci, the directrices, the vertices, and the asymptotes of

$$\frac{y^2}{9} - \frac{x^2}{7} = 1.$$

Answer

Here $b = 3$ and $a = \sqrt{7}$. Hence

$$e = \sqrt{\frac{a^2 + b^2}{b^2}} = \sqrt{\frac{7 + 9}{9}} = \frac{4}{3},$$

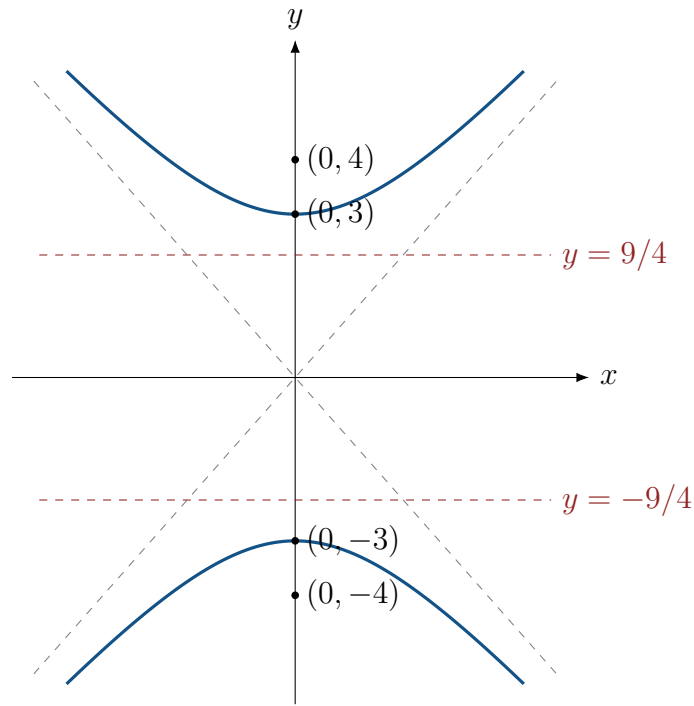
centre = $(0, 0)$,

foci = $(0, \pm be) = (0, \pm 4)$,

vertices = $(0, \pm b) = (0, \pm 3)$,

directrices : $y = \pm \frac{b}{e} = \pm \frac{9}{4}$,

asymptotes : $y = \pm \frac{b}{a}x = \pm \frac{3}{\sqrt{7}}x$.



Example 6: A Translated Horizontal Hyperbola

Sketch the hyperbola and label its vertices and foci. Also find the equations of the directrices and asymptotes:

$$16x^2 - y^2 - 32x - 6y = 57.$$

Answer

Complete the squares:

$$\begin{aligned} 16(x^2 - 2x) - (y^2 + 6y) &= 57, \\ 16((x - 1)^2 - 1) - ((y + 3)^2 - 9) &= 57, \\ 16(x - 1)^2 - (y + 3)^2 &= 64, \\ \frac{(x - 1)^2}{4} - \frac{(y + 3)^2}{64} &= 1. \end{aligned}$$

Thus $a = 2$, $b = 8$, and the centre is $(1, -3)$. Also,

$$\begin{aligned} e &= \sqrt{\frac{a^2 + b^2}{a^2}} = \sqrt{\frac{4 + 64}{4}} = \sqrt{17}, \\ c &= ae = 2\sqrt{17}, \\ \text{vertices} &= (1 \pm 2, -3) = (-1, -3), (3, -3), \\ \text{foci} &= (1 \pm 2\sqrt{17}, -3), \\ \text{directrices} : x &= 1 \pm \frac{a}{e} = 1 \pm \frac{2}{\sqrt{17}}, \\ \text{asymptotes} : y + 3 &= \pm \frac{b}{a}(x - 1) = \pm 4(x - 1). \end{aligned}$$

